A Dense Fixture

reliable systems fail at the seams where two correct components meet and disagree

every retry needs a budget because unbounded retries are just outages in disguise

a queue is a promise that work happens later, never proof that it happens at all

caching is the art of deciding which stale answer is still true enough to serve

timeouts are a design decision about how long you are willing to wait for bad news

backpressure is what you owe downstream when you cannot say no upstream

an idempotency key turns a scary repeat into a boring no-op at the sink

the fallback path is production the moment the primary path has a bad day

observability is paying attention in production instead of hoping in staging

degrade on purpose rather than by accident, and say which parts bend first

small batch sizes turn a catastrophe into a changelog entry nobody reads

feature flags are how you ship code without shipping the behavior on day one

the health check must ask a question the service can actually answer

capacity planning is guessing honestly, in writing, with numbers attached

rollbacks are a feature you buy before the incident, never during it

every alert should page a human only when a human can change the outcome